

LEFT OUTER JOIN

Also called simply LEFT JOIN. The keyword OUTER is optional.

```
SELECT ...  
FROM A  
LEFT JOIN B  
ON condition;
```

15.4.1 Definition

A LEFT JOIN returns:

- ✓ Every row from the left table
- ✓ Matching rows from the right table
- ✓ NULL for right-table columns when no match exists

Think of it as:

"Keep everything on the left. If the right side matches, bring it. Otherwise fill the right side with NULL."

Example

Students

STUDENT_ID	NAME
1	Arpit
2	Aadz
3	Riya

Enrollments

STUDENT_ID	COURSE
1	DBMS
2	OS

Query

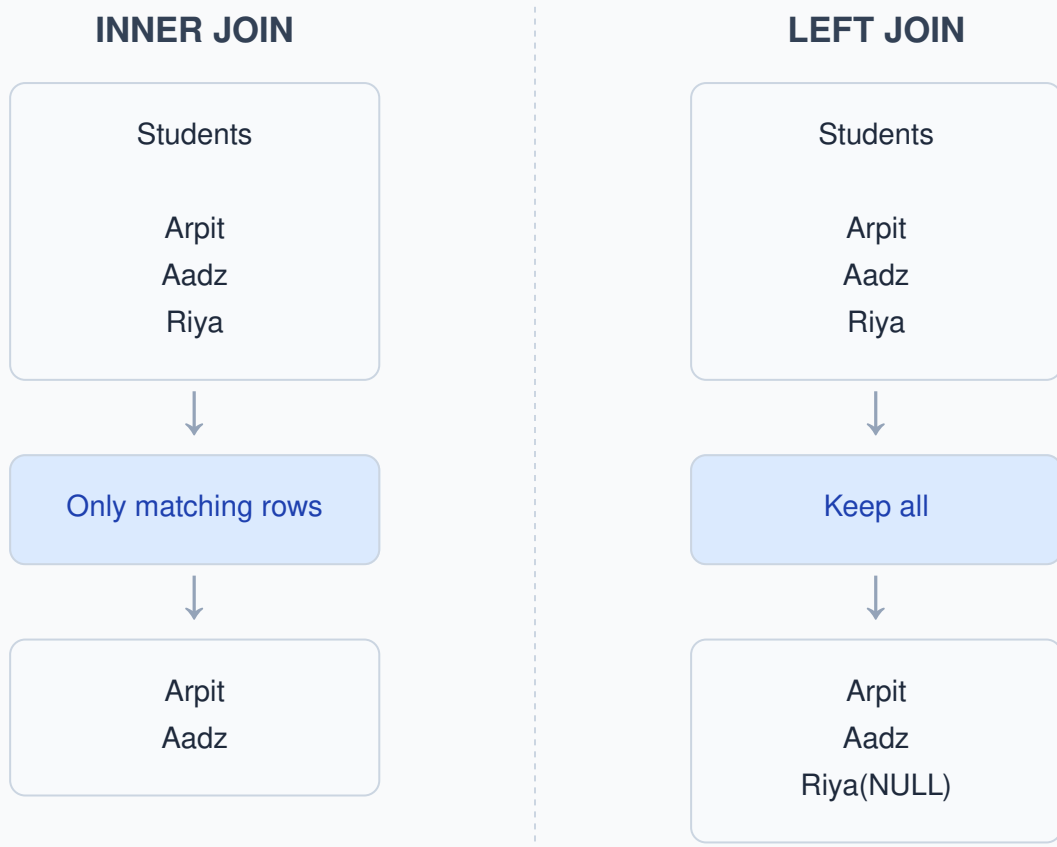
```
SELECT
    s.name,
    e.course
FROM Students s
LEFT JOIN Enrollments e
ON s.student_id = e.student_id;
```

Output

NAME	COURSE
Arpit	DBMS
Aadz	OS
Riya	NULL

Notice Riya is still present.

INNER JOIN vs LEFT JOIN



Internal Execution

SQL conceptually performs:



Step 4

For unmatched left rows

Generate

Left row + NULL NULL NULL

This automatic NULL generation is what makes LEFT JOIN special.

Multiple Matches

Students

ID	NAME
1	Arpit

Enrollments

STUDENT_ID	COURSE
1	DBMS
1	OS

Output

NAME	COURSE
Arpit	DBMS
Arpit	OS

Still duplicates.

LEFT JOIN never removes duplicates.

| ON vs WHERE (Most Important Interview Topic)

Students

ID	NAME
1	Arpit
2	Aadz

Enrollments

STUDENT_ID	COURSE
1	DBMS

Query 1

```
SELECT *  
FROM Students s  
LEFT JOIN Enrollments e  
ON s.student_id=e.student_id;
```

Output

NAME	COURSE
Arpit	DBMS
Aadz	NULL

Now

```
SELECT *  
FROM Students s  
LEFT JOIN Enrollments e  
ON s.student_id=e.student_id  
WHERE e.course='DBMS';
```

Output

NAME	COURSE
Arpit	DBMS

Aadz disappears.

Why?

After LEFT JOIN

Aadz NULL

WHERE

NULL= 'DBMS '

FALSE

Row removed.

Your LEFT JOIN effectively became an INNER JOIN.

Correct filtering

```
SELECT *  
FROM Students s  
LEFT JOIN Enrollments e  
ON s.student_id=e.student_id  
AND e.course='DBMS';
```

Now Output

NAME	COURSE
Arpit	DBMS
Aadz	NULL

Huge interview favorite.

Multiple LEFT JOINS

Students

↓ LEFT JOIN Enrollment

↓ LEFT JOIN Courses

↓ LEFT JOIN Teachers

Every unmatched stage continues producing NULLs.

Problem / Why

PROBLEM	WHY
175. Combine Two Tables	Classic LEFT JOIN
577. Employee Bonus	LEFT JOIN + NULL
1581. Customer Who Visited but Did Not Make Any Transactions	LEFT JOIN + IS NULL
570. Managers with at Least 5 Direct Reports	LEFT JOIN understanding helps

Example Solution — LeetCode 175

```
SELECT
    p.firstName,
    p.lastName,
    a.city,
    a.state
FROM Person p
LEFT JOIN Address a
ON p.personId=a.personId;
```

Example Solution — LeetCode 577

```
SELECT
    e.name,
    b.bonus
FROM Employee e
LEFT JOIN Bonus b
ON e.empId=b.empId
WHERE b.bonus < 1000
    OR b.bonus IS NULL;
```

Example Solution — LeetCode 1581

```
SELECT
    v.customer_id,
    COUNT(*) AS count_no_trans
FROM Visits v
LEFT JOIN Transactions t
ON v.visit_id=t.visit_id
WHERE t.transaction_id IS NULL
GROUP BY v.customer_id;
```

This is one of the most frequently asked LEFT JOIN patterns.

RIGHT OUTER JOIN

Syntax

```
SELECT ...  
FROM A  
RIGHT JOIN B  
ON ...;
```

Definition

Keep

- every row from the right table
- matching rows from the left
- NULL when left has no match

Example

Students

ID	NAME
1	Arpit

Enrollments

STUDENT_ID	COURSE
1	DBMS
2	OS

Query

```
SELECT
    s.name,
    e.course
FROM Students s
RIGHT JOIN Enrollments e
ON s.id=e.student_id;
```

Output

NAME	COURSE
Arpit	DBMS
NULL	OS

| Should You Use RIGHT JOIN?

In industry

Almost never.

Instead of

```
A
RIGHT JOIN
B
```

people simply write

```
B
LEFT JOIN
A
```

Same result.

Much easier to read.

ULL OUTER JOIN

Definition

Return

- matching rows
- left-only rows
- right-only rows

Everything.

Example

Students

ID	NAME
1	Arpit
2	Aadz

Enrollments

STUDENT_ID	COURSE
2	DBMS
3	OS

Output

NAME	COURSE
Arpit	NULL
Aadz	DBMS
NULL	OS

| MySQL Doesn't Support FULL OUTER JOIN

Classic interview question.

Simulate using

```
SELECT *  
FROM A  
LEFT JOIN B  
ON A.id=B.id  
  
UNION  
  
SELECT *  
FROM A  
RIGHT JOIN B  
ON A.id=B.id;
```

Or, more commonly:

```
SELECT *  
FROM A  
LEFT JOIN B  
ON A.id = B.id  
  
UNION  
  
SELECT *  
FROM B  
LEFT JOIN A  
ON A.id = B.id;
```

UNION removes duplicate matching rows, giving the effect of a FULL OUTER JOIN

Employee

EMPLOYEE_ID	NAME
2	Arpit
4	Aadz

Salary

EMPLOYEE_ID	SALARY
2	5000
3	7000

Need employees that appear in only one table.

Output

EMPLOYEE_ID
3
4

```
SELECT e.employee_id
FROM Employees e
LEFT JOIN Salaries s
ON e.employee_id=s.employee_id
WHERE s.employee_id IS NULL

UNION

SELECT s.employee_id
FROM Salaries s
LEFT JOIN Employees e
ON s.employee_id=e.employee_id
WHERE e.employee_id IS NULL
ORDER BY employee_id;
```